

## Practical - 3

**Practical Title:** Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

**Problem Definition:** Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.

### A. Theory

#### 1. How does the layout adapt for mobile, tablet, and desktop screens?

**Answer:**

The StudentHub layout is designed using a mobile-first approach. The basic layout is created for smaller screens first, and then CSS media queries are used to adjust the layout for tablets and desktops.

On mobile screens, content is arranged mostly in a single column so that it remains easy to read. On tablets, elements can be arranged into multiple columns where space is available. On desktop screens, wider layouts, grids, and multiple content sections can be displayed.

This approach allows the same website to work properly on different screen sizes without creating separate websites for each device.

**Reference:**

MDN Web Docs – Responsive design

[https://developer.mozilla.org/en-US/docs/Learn\\_web\\_development/Core/CSS\\_layout/Responsive\\_Design](https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Responsive_Design)

---

#### 2. Which layout technique is used and why?

**Answer:**

Both CSS Flexbox and CSS Grid are used in the StudentHub project.

Flexbox is mainly used for one-dimensional layouts such as navigation bars, buttons, cards, and arranging elements horizontally or vertically.

CSS Grid is used when a two-dimensional layout is required, such as displaying multiple cards, dashboard sections, events, or other content in rows and columns.

Using both techniques makes the layout easier to organize and responsive across different screen sizes.

**Reference:**

MDN Web Docs – CSS Flexbox

[https://developer.mozilla.org/en-US/docs/Learn\\_web\\_development/Core/CSS\\_layout/Flexbox](https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Flexbox)

MDN Web Docs – CSS Grid

[https://developer.mozilla.org/en-US/docs/Learn\\_web\\_development/Core/CSS\\_layout/Grids](https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Grids)

---

### **3. Are typography, spacing, colors, and contrast readable?**

**Answer:**

Yes. Typography, spacing, colors, and contrast are selected to make the StudentHub interface easy to read and navigate.

Readable font sizes are used for headings and normal text. Proper spacing is maintained between sections, cards, buttons, and form elements. The color combination provides sufficient contrast between the background and text. Important buttons and interactive elements are also visually distinguishable.

The same design system is maintained across the different pages to provide a consistent user experience.

**Reference:**

MDN Web Docs – CSS styling

[https://developer.mozilla.org/en-US/docs/Learn\\_web\\_development/Core/Styling\\_basics](https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Styling_basics)

---

### **4. Is the framework used consistently without unnecessary inline CSS?**

**Answer:**

Yes. If Bootstrap is used in the StudentHub project, its components and utility classes are used consistently wherever they are useful.

Custom CSS is used for project-specific styling that is not provided by the framework. Unnecessary inline CSS is avoided because keeping styles in CSS files makes the project easier to maintain and update.

The framework and custom CSS are organized so that the design remains consistent across all pages.

**Reference:**

Bootstrap Documentation

<https://getbootstrap.com/docs/>

---

## **B. Implementation/ Output/Result**

The StudentHub portal was designed using **HTML5, CSS3, Flexbox, CSS Grid, and responsive media queries**.

Responsive layouts were implemented for the following pages:

1. Home
2. About
3. Registration
4. Dashboard
5. Events

### **Example Responsive CSS**

#### **Result**

The StudentHub pages were successfully designed with responsive layouts. CSS Grid was used for structured content layouts, while Flexbox was used for navigation and one-dimensional arrangements.

The interface adapts to mobile, tablet, and desktop screen sizes while maintaining consistent typography, spacing, colors, and overall design.

---

## **C. Evidence of Work (Screenshots)**

## Student Login

**Student ID**

**Password**

Login

## Student Registration

Full Name

Student ID

Password

Register

Already have an account? [Login](#)

 **Student Hub Portal**

Home | Profile | Attendance | Courses | Results | Timetable | Contact | Logout

## Welcome to Student Dashboard

Welcome to the Student Hub Portal.

### Student Information

Name : Rehan Saiyed  
Student ID : 24CE001  
Department : Computer Engineering  
Semester : III

### Notice Board

 **Student Hub Portal**

Home | Profile | Attendance | Courses | Results | Timetable | Contact | Logout

## Profile

| Field       | Details              |
|-------------|----------------------|
| Name        | Rehan Saiyed         |
| Student ID  | 24CE001              |
| Department  | Computer Engineering |
| Semester    | III                  |
| Division    | A                    |
| Email       | rehan@example.com    |
| Phone       | 9876543210           |
| Blood Group | B+                   |

Student Hub Portal

Home | Profile | Attendance | Courses | Results | Timetable | Contact | Logout

### Welcome to Student Dashboard

Welcome to the Student Hub Portal.

#### Student Information

Name : Rehan Saiyed  
Student ID : 24CE001  
Department : Computer Engineering

 Snipping Tool

Screenshot copied to clipboard  
Automatically saved to screenshots folder.

Mark-up and share



# Student Hub Portal

[Home](#) | [Profile](#) | [Attendance](#) | [Courses](#) | [Results](#) | [Timetable](#) | [Contact](#) | [Logout](#)

## Attendance Record

| Subject | Total Classes | Attended | Attendance |
|---------|---------------|----------|------------|
| WDF     | 30            | 28       | 93%        |
| FDSA    | 32            | 30       | 94%        |
| FCN     | 28            | 25       | 89%        |
| OOD     | 26            | 24       | 92%        |



## Student Hub Portal

[Home](#) | [Profile](#) | [Attendance](#) | [Courses](#) | [Results](#) | [Timetable](#) | [Contact](#) | [Logout](#)

### Profile

| Field       | Details              |
|-------------|----------------------|
| Name        | Rehan Saiyed         |
| Student ID  | 24CE001              |
| Department  | Computer Engineering |
| Semester    | III                  |
| Division    | A                    |
| Email       | rehan@example.com    |
| Phone       | 9876543210           |
| Blood Group | B+                   |

### Notice Board

 **Student Hub Portal**
[Home](#) | [Profile](#) | [Attendance](#) | [Courses](#) | [Results](#) | [Timetable](#) | [Contact](#) | [Logout](#)
**Attendance Record**

| Subject | Total Classes | Attended | Attendance |
|---------|---------------|----------|------------|
| WDF     | 30            | 28       | 93%        |
| FDSA    | 32            | 30       | 94%        |
| FCN     | 28            | 25       | 89%        |
| OOP     | 26            | 24       | 92%        |
| DM      | 30            | 27       | 90%        |
| CPI     | 20            | 19       | 95%        |

Overall Attendance: 92%

**Notice Board**

Assignment Submission : Friday

Mid Semester Exam : Next Month

 **Student Hub Portal**
[Home](#) | [Profile](#) | [Attendance](#) | [Courses](#) | [Results](#) | [Timetable](#) | [Contact](#) | [Logout](#)
**courses details****Enrolled Courses**

| Course Code | Course Name                                    | Faculty            |
|-------------|------------------------------------------------|--------------------|
| CEUC201     | Fundamentals of Data Structures and Algorithms | nikita bhatt       |
| CEUC202     | Object Oriented Programming                    | deven gol          |
| ITUC201     | Fundamentals of Computer Networks              | nishant sir        |
| MSUD203     | Discrete Mathematics                           | rajesh v. savaliya |
| CEUE202     | Web Development Frameworks                     | sachin sir         |
| HSUV201     | Creativity, Problem Solving and Innovation     | vikash sir         |

**Notice Board**



## Student Hub Portal

[Home](#) | [Profile](#) | [Attendance](#) | [Courses](#) | [Results](#) | [Timetable](#) | [Contact](#) | [Logout](#)

### student result

#### Semester Result

| Subject              | Marks | Grade |
|----------------------|-------|-------|
| Web Development      | 88    | A     |
| Java Programming     | 82    | A     |
| Data Structures      | 79    | B+    |
| Digital Electronics  | 85    | A     |
| Discrete Mathematics | 90    | A+    |

SGPA : 8.48

Student ID : 25DCE191



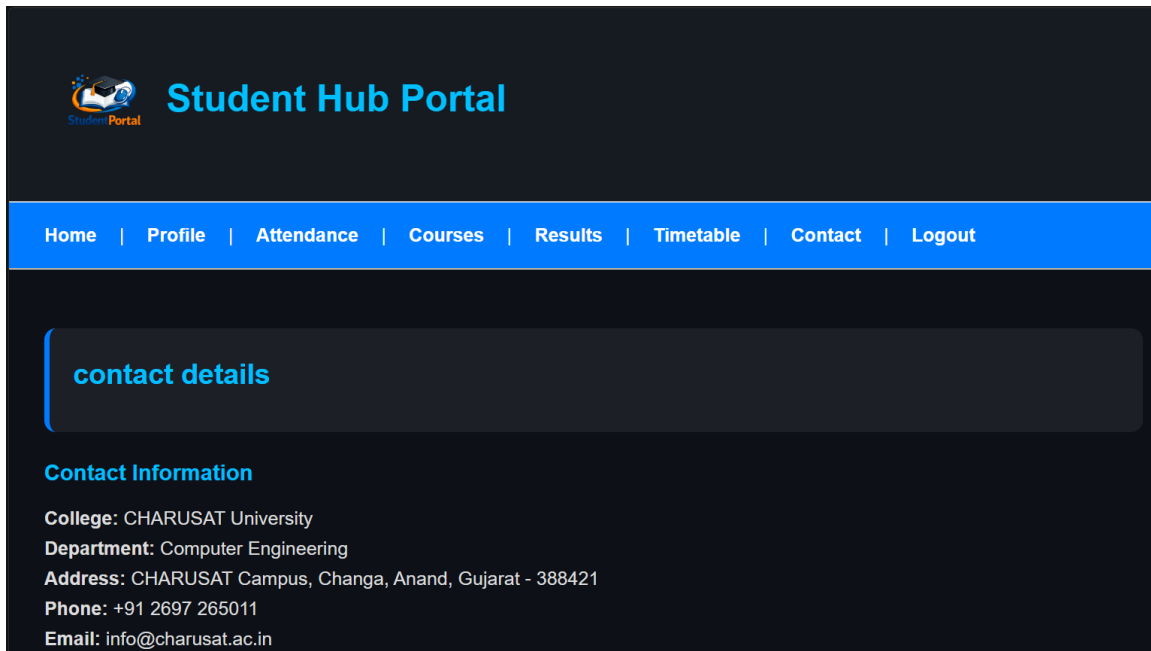
## Student Hub Portal

[Home](#) | [Profile](#) | [Attendance](#) | [Courses](#) | [Results](#) | [Timetable](#) | [Contact](#) | [Logout](#)

### time table

#### Class Timetable

| Day       | 9:10 AM - 11:10 AM | 11:10 AM - 12:10 PM | 12:10 PM - 2:10 PM      | 2:10 PM - 2:20 PM | 2:20 PM - 4:20 PM |
|-----------|--------------------|---------------------|-------------------------|-------------------|-------------------|
| Monday    | DM (T)<br>FCN (T)  | Break               | OOP (P)                 | Short Break       | CPI (T)           |
| Tuesday   | DM (T)<br>FDSA (T) | Break               | Counselling / Mentoring | Short Break       | FDSA (P)          |
| Wednesday | WDF (P)            | Break               | FCN (T)<br>Counselling  | Short Break       | FCN (T)<br>DM (T) |
| Thursday  | WDF (T)            | Break               | FDSA (T)                | Short Break       | FCN (P)           |



The screenshot shows the StudentHub dashboard in mobile responsive mode. Content is rearranged into a smaller and easier-to-read layout suitable for a narrow screen.

---

#### D. GitHub Repository Link:

**GitHub Repository:** <https://github.com/rehansaiyed72/student-hub-portal>

---

#### E. Challenges Faced:

During the implementation of the responsive StudentHub interface, the following challenges were faced:

1. Understanding how CSS Grid and Flexbox should be used for different layouts.
2. Making the same page work correctly on different screen sizes.
3. Adjusting the number of columns for cards and dashboard sections.
4. Preventing text, buttons, and navigation elements from overflowing on mobile screens.
5. Maintaining consistent spacing and typography across different devices.

6. Using media queries correctly for tablet and mobile breakpoints.
7. Keeping custom CSS organized without depending too much on inline styling.

These challenges were solved by testing the pages using the browser's responsive design mode and adjusting the Grid, Flexbox, spacing, and media-query rules accordingly.

---

#### **F. Learning Outcomes:**

After completing this practical, I learned:

1. How to create responsive web layouts using CSS3.
2. How to use Flexbox for arranging elements in one direction.
3. How to use CSS Grid for row and column-based layouts.
4. How media queries help websites adapt to different screen sizes.
5. How to follow a mobile-first approach when designing webpages.
6. How to maintain consistent typography, spacing, colors, and contrast.
7. How to test responsive webpages using browser developer tools.
8. How to organize CSS so that it can be reused across multiple pages.
9. How responsive design improves the usability of websites on different devices.

#### **Faculty Remarks:**